10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

# Exercise 5.1 — Reducing to Lowest Form

Unit 5 | Solved Exercise | New Book (PECTAA)

## Exercise 5.1 (Solved)

To reduce an algebraic fraction to its lowest form we factorise the numerator and the denominator, then cancel the common factors. The value of the fraction does not change by cancelling a common factor.

### 1. Reduce the following rational expressions to lowest forms.

**(i)** $\dfrac{3a^2 - 6ab}{2a^2b - 4ab^2}$

Take out the common factors in the numerator and the denominator

$$\frac{3a^2 - 6ab}{2a^2b - 4ab^2} = \frac{3a(a - 2b)}{2ab(a - 2b)}$$

Cancel the common factor $(a - 2b)$

$$= \frac{3a}{2ab}$$

Cancel the common factor $a$

$$= \frac{3}{2b}$$

**Answer: 3/(2b)**

**(ii)** $\dfrac{abx + bx^2}{acx + cx^2}$

Take out $bx$ from the numerator and $cx$ from the denominator

$$\frac{abx + bx^2}{acx + cx^2} = \frac{bx(a + x)}{cx(a + x)}$$

Cancel the common factor $(a + x)$

$$= \frac{bx}{cx}$$

Cancel the common factor $x$

$$= \frac{b}{c}$$

**Answer: b/c**

**(iii)** $\dfrac{ac}{a^2x^2 - ax}$

Take out $ax$ from the denominator

$$\frac{ac}{a^2x^2 - ax} = \frac{ac}{ax(ax - 1)}$$

Cancel the common factor $a$

$$= \frac{c}{x(ax - 1)}$$

**Answer: c/[x(ax − 1)]**

studysapiens.com

**(iv)** $\frac{15a^2b^2c}{100(a^2 - a^2b)}$

Take out $a^2$ from the bracket of the denominator

$\frac{15a^2b^2c}{100(a^2 - a^2b)}$

$= \frac{15a^2b^2c}{100a^2(1 - b)}$

Cancel $a^2$

$= \frac{15b^2c}{100(1 - b)}$

Divide 15 and 100 by 5

$= \frac{3b^2c}{20(1 - b)}$

**Answer: $3b^2c/[20(1 - b)]$**

**(v)** $\frac{4x^2 - 9y^2}{4x^2 + 6xy}$

Write the numerator as a difference of two squares and take 2x out of the denominator

$\frac{4x^2 - 9y^2}{4x^2 + 6xy}$

$= \frac{(2x - 3y)(2x + 3y)}{2x(2x + 3y)}$

Cancel the common factor (2x + 3y)

$= \frac{2x - 3y}{2x}$

**Answer: (2x − 3y)/(2x)**

**(vi)** $\frac{20(x^3 - y^3)}{5x^2 + 5xy + 5y^2}$

Take 5 out of the denominator and write $x^3 - y^3$ as $(x - y)(x^2 + xy + y^2)$

$\frac{20(x^3 - y^3)}{5x^2 + 5xy + 5y^2}$

$= \frac{20(x - y)(x^2 + xy + y^2)}{5(x^2 + xy + y^2)}$

Cancel the common factor $(x^2 + xy + y^2)$

$= \frac{20(x - y)}{5}$

Divide 20 and 5 by 5

$= 4(x - y)$

**Answer: 4(x − y)**

**(vii)** $\frac{x(2a^2 - 3ax)}{a(4a^2x - 9x^3)}$

Take out a from the numerator, and ax from the denominator

$\frac{x(2a^2 - 3ax)}{a(4a^2x - 9x^3)}$

$= \frac{xa(2a - 3x)}{ax(4a^2 - 9x^2)}$

Write $4a^2 - 9x^2$ as $(2a - 3x)(2a + 3x)$

$= \frac{xa(2a - 3x)}{ax(2a - 3x)(2a + 3x)}$

Cancel the common factor ax(2a − 3x)

$= \frac{1}{2a + 3x}$

studysapiens.com

**Answer: 1/(2a + 3x)**

**(viii)** $\frac{x^2 - 5x}{x^2 - 4x - 5}$

Take x out of the numerator and factorise the denominator

$$\frac{x^2 - 5x}{x^2 - 4x - 5} = \frac{x(x - 5)}{(x - 5)(x + 1)}$$

Cancel the common factor (x − 5)

$$= \frac{x}{x + 1}$$

**Answer: x/(x + 1)**

**(ix)** $\frac{3x^2 + 6x}{x^2 + 4x + 4}$

Take 3x out of the numerator and write the denominator as a perfect square

$$\frac{3x^2 + 6x}{x^2 + 4x + 4} = \frac{3x(x + 2)}{(x + 2)^2}$$

Cancel one factor (x + 2)

$$= \frac{3x}{x + 2}$$

**Answer: 3x/(x + 2)**

**(x)** $\frac{x^2 + xy - 2y^2}{x^3 - y^3}$

Factorise the numerator and write $x^3 - y^3$ as $(x - y)(x^2 + xy + y^2)$

$$\frac{x^2 + xy - 2y^2}{x^3 - y^3} = \frac{(x + 2y)(x - y)}{(x - y)(x^2 + xy + y^2)}$$

Cancel the common factor (x − y)

$$= \frac{x + 2y}{x^2 + xy + y^2}$$

**Answer: $(x + 2y)/(x^2 + xy + y^2)$**

**(xi)** $\frac{2x^2 + 17x + 21}{3x^2 + 26x + 35}$

Factorise both the quadratic expressions

$2x^2 + 17x + 21 = (2x + 3)(x + 7)$

$3x^2 + 26x + 35 = (3x + 5)(x + 7)$

$$\frac{2x^2 + 17x + 21}{3x^2 + 26x + 35} = \frac{(2x + 3)(x + 7)}{(3x + 5)(x + 7)}$$

Cancel the common factor (x + 7)

$$= \frac{2x + 3}{3x + 5}$$

**Answer: (2x + 3)/(3x + 5)**

Remember: only factors can be cancelled, never the separate terms of a sum. That is why we factorise first and cancel afterwards.

studysapiens.com